

Understanding the Mechanism of Drug Resistance Due to a Codon Deletion in Protoporphyrinogen Oxidase through Computational Modeling

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Protoporphyrinogen oxidase (PPO; EC 1.3.3.4) is the last common enzyme for the enzymatic transformation of protoporphyrinogen-IX to protoporphyrin-IX, which is the key common intermediate leading to heme and chlorophyll. Hence, PPO has been identified as one of the most importance action targets for the treatment of some important diseases including cancer and variegated porphyria (VP). In the agricultural field, PPO inhibitors have been used as herbicides for many years. Recently, a unique drug resistance was found to be associated with a nonactive site residue (Gly210) deletion rather than substitution in *A. tuberculatus* PPO. In the present study, extensive computational simulations, including homology modeling, molecular dynamics (MD) simulations, and molecular mechanics-Poisson–Boltzmann surface area (MM-PBSA) calculations, have been carried out to uncover the detailed molecular mechanism of drug resistance associated with Gly210 deletion. Although Gly210 in the wild-type *A. tuberculatus* PPO has no direct interaction with the inhibitors, all the computational models and energetic results indicated that Gly210 deletion has great effects on the hydrogen-bonding network and the conformational change of the binding pocket. An interchain hydrogen bond between Gly210 with Ser424, playing an important role in stabilizing the local conformation of the wild-type enzyme, disappeared after Gly210 deletion. As a result, the mutant-type PPO has a lower affinity than the wild-type enzyme, which accounts for the molecular mechanism of drug resistance. The structural and mechanistic insights obtained from the present study provide a new starting point for future rational design of novel PPO inhibitors to overcome drug resistance associated with Gly210 deletion.

Introduction

Protoporphyrinogen-IX oxidase (PPO; EC 1.3.3.4), the last common enzyme in the biosynthetic pathway leading to heme and chlorophyll synthesis, catalyzes the oxidation of protoporphyrinogen-IX to protoporphyrin-IX by molecular oxygen.^{1–4} In mammals, there is only one isoform of protoporphyrinogen-IX oxidase located on the cytosolic side of the inner mitochondrial membrane,^{4–7} whereas in plants there are two isoforms, i.e., the plastidic PPO (PPO1) and the mitochondrial PPO (PPO2).^{8–12} Although the oxidation of protoporphyrinogen-IX to protoporphyrin-IX can take place nonenzymatically, the presence of PPO that catalyzes this reaction seems to be ubiquitous in cells having aerobic metabolism.^{2,13} Inhibitors of this enzyme do more than merely blocking the production of heme and chlorophyll. When the enzyme is inhibited, the substrate protoporphyrinogen-IX will accumulate and be exported to the cytoplasm where it is slowly oxidized by O₂ in the mitochondrion and chloroplast, producing protoporphyrin IX. This spontaneous production may bypass the inhibitor and allow the cell to function normally, but it has dire consequences: In the presence of light, the photosensitive protoporphyrin IX generates singlet oxygen that causes lipid peroxidation and cell death.^{14–18} Interest in the protoporphyrinogen oxidase arises

from both its unique biochemical characteristics and its biomedical significance.

The biomedical significance of protoporphyrinogen-IX oxidase can be concluded in three different aspects. For one of the reasons, partial PPO deficiency in humans causes an inherited disease known as variegated porphyria (VP) characterized by cutaneous photosensitivity and the propensity to develop acute neurovisceral crisis.^{19–23} This disease is relatively common in the white African population of South Africa.^{24,25} The disease is most common in the young adults but may suddenly occur at any age including the adolescent and elderly.^{24,26} It is now acknowledged that the PPO deficiency is the primary genetic defect in VP; the defective gene is located on chromosome number 1 and is inherited in an autosomal dominant fashion.^{27–29} In patients with this disease, the activity of PPO is reduced by at least 50%.³⁰ The symptomology of VP and its highly variable penetrance of infected individuals make the study of the nature of PPO causing the disease of great interest.

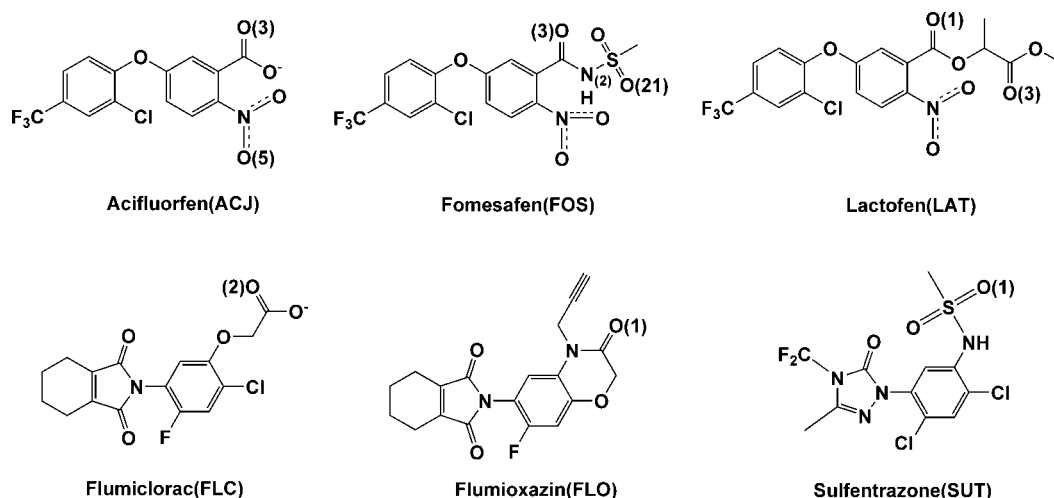
Another reason for the increasing interest in protoporphyrinogen oxidase is that in plants this enzyme has been found to be a significant target for a number of herbicides such as diphenylethers, phenylpyrazoles, oxadiazoles, triazolinones, thiadiazoles, pyrimidinones, oxazolidinedione, and *N*-phenylphthalimides that have been produced commercially as herbicides for many years.^{15–34} These chemicals inhibit protoporphyrinogen oxidase, resulting in cytoplasmic accumulation of the substrate protoporphyrinogen-IX, which is a highly efficient photosensitizer leading to severe photochemical damage

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SCHEME 1: Chemical Structures of Some Representative PPO Inhibitors



when treated plants are exposed to visible radiation. Therefore, PPO-inhibiting herbicides are known as light-dependent bleaching herbicides.³²

In addition, a potentially important medical application of PPO inhibitors is associated with photodynamic therapy (PDT).^{35,36} PDT has been used in the detection and treatment of cancer and other diseases and is also potentially valuable in destroying bacteria and other dangerous organisms.³⁷ PDT matured as a feasible medical technology in the 1980s at several institutions throughout the world as a third-level treatment involving three key components: a photosensitizer, light, and tissue oxygen.^{38–44} For example, in the first step of PDT for cancer treatment, a photosensitizing agent is injected into the bloodstream and then absorbed by cells all over the body, but it stays in cancer cells longer than it does in normal cells. Approximately 24–72 h after injection,³⁸ when most of the agent has left normal cells but remains in cancer cells, the tumor is exposed to light. The photosensitizer in the tumor absorbs the light and produces an active form of oxygen that destroys nearby cancer cells.^{38–40} The most potentially useful application of PPO inhibitors is a treatment for destroying cancer cells via PDT. If photosensitizers are established in high concentrations in the tumor cells and these tumors are exposed to irradiation with light of an appropriate wavelength, the photosensitizer will be activated and destroy the cells.⁴⁵ Protoporphyrinogen-IX is an extremely effective photosensitizer, but it is not useful before activation. Halling et al.⁴⁵ showed that PPO inhibitors could activate the photosensitizer protoporphyrinogen-IX and cause its accumulation within tumor cells. The practical use of PPO inhibitors in PDT is being explored further.

PPO inhibitors provide a useful tool to investigate the substrate-binding site and catalytic mechanism of the enzyme. They are competitive inhibitors with respect to the tetrapyrrole substrate protoporphyrinogen-IX of PPO in plants, mice, and humans. However, one of the most serious concerns about PPO is the mutation which could result in decreased activity to the substrate and simultaneously cause resistance to PPO inhibitors.^{28–30} Most mutations (for example, R59W, R168C, G232R, and Y348C) causing a high level of decreased activity occur at the conserved residue of human PPO. Further, heterogeneity of PPO variants has been reported worldwide with over 100 mutations identified.^{28–30} Meanwhile, engineered overproduction of PPO in plants may confer resistance toward these herbicides. To date, the naturally evolved resistance to PPO-inhibiting herbicides has been reported for three types of weeds.⁴⁶ Generally speaking,

the usual mechanism of resistance is the amino acid substitution within or nearby the active site. However, an incredible and unique resistance mechanism, i.e., an amino acid deletion rather than a substitution, was recently found to be associated with PPO inhibitor-resistant *Amaranthus tuberculatus* (waterhemp) biotype.^{47,48} It should be noted that the deleted residue (Gly210) in the native enzyme is not in the active site and, therefore, is not expected to have a direct interaction with the substrate or the known inhibitors.⁴⁷ An extremely interesting observation reported is that the Gly210 deletion does not decrease the catalytic activity of PPO for its substrate, but it does cause resistance to almost all PPO inhibitors. Although many examples about mutations via amino acid residue substitutions have been reported to cause a high level of drug resistance,^{49–52} the unique mechanism due to amino acid residue deletion has not yet been elucidated. It is of great interest to understand the mechanism at the molecular level of why a residue deletion confers broad drug resistance to the inhibitors when the same residue deletion does not change the catalytic activity of the enzyme for its substrate.

In this study, various computational techniques, including homology modeling, molecular docking, molecular dynamics (MD) simulations, and the molecular mechanics-Poisson-Boltzmann surface area (MM-PBSA) calculations, were used to uncover the detailed mechanism of the drug resistance due to the codon deletion. Some commercial PPO inhibitors, such as acifluorfen (ACF), fomesafen (FOS), lactofen (LAT), flumiclorac (FLC), flumioxazin (FLO), and sulfentrazone (SUT) depicted in Scheme 1, were considered as typical examples in the present study. The computational results reveal that the conformational change of the binding pocket and the hydrogen-bonding network change caused by the residue deletion are responsible for the decrease of the ligand-binding affinity. Our findings will provide a new starting point for rational design of novel PPO inhibitors to overcome the drug resistance associated with Gly210 deletion.

Computational Methods

Homology Modeling. To understand the resistance mechanism due to a codon deletion, first the 3D structure of the enzyme must be known. Although the X-ray crystal structure of the tobacco (*Nicotiana tabacum*) and *Myxococcus Xanthus* PPO was reported, this kind of mutation was only found in *A. tuberculatus*, and some experimental data were available. The

combined use of X-ray crystal structure of the tobacco PPO2 and the computational chemistry approach enables a reasonable 3D structure of *A. tuberculatus* to be built. Homology modeling is usually the method of choice when a clear relationship of homology between the sequence of target protein and at least one known structure is found. This approach can give reasonable results based on the assumption that the tertiary structures of two proteins are similar if their sequences are related.⁵³ The sequence of *A. tuberculatus* PPO2 (DQ386117) was obtained from the nucleotide data bank of National Center for Biotechnology Information (NCBI: <http://www.ncbi.nlm.nih.gov>). Although the crystal structure of *M. Xanthus* PPO includes the ligand ACJ, it has low sequence similarity with the model sequence. The X-ray crystal structure of tobacco PPO2 (PDB entry of 1SEZ at 2.9 Å resolution) was used as a reference protein. The solved tobacco PPO2 structure includes two equivalent monomers. Only one of these equivalent monomers is needed to be considered because their equivalent binding sites are largely separated, and thus the ligand binding in one site should have no significant effect on the other ligand binding in the other binding site. The sequences of DQ386117 and tobacco PPO2 were aligned using the program ClustalW (<http://www.ebi.ac.uk/clustalw>). The DQ386117 has 67% sequence identity with tobacco PPO2. The sequence alignment result of tobacco PPO2 and DQ386117 is shown in the Supporting Information. To study PPO2 inhibitor-resistance mechanism in atomic detail, a homology model of the *A. tuberculatus* PPO2 was built using the Modeler module of InsightII (version 2005) developed by Accelrys Software Inc.^{54,55}

The initial structure of modeling was revised first by means of refining loops and rotamers, checking bonds, and adding hydrogen atoms. Then, the protein structure was energy-minimized by using the Discover module of InsightII for 2000 steps of steepest descent optimization calculations and 5000 steps of conjugated gradient optimization calculations. The fully energy-minimized structure was carefully checked to make sure that the overall structure of the final model was not significantly different from that of the template. Finally, the modeled structure was validated by using the PROCHECK program,⁵⁶ molecular dynamics (MD) simulation, and other different protocols (see below).

Molecular Docking. After the final model of the *A. tuberculatus* PPO2 was obtained, the GOLD program^{57,58} was used to dock the molecular species of ACJ, FOS, LAT, FLC, FLO, and SUT into the binding site. An active site structure with a radius of 10 Å was defined considering a molecular reference as the center. The atomic charges used for all the ligands were the Gasteiger–Huckel charges. GOLD docking was performed using default parameters, automatic settings with 100% search efficiency.⁵⁹ Finally, the enzyme–ligand complex structures were selected according to the criteria for GOLDScore combined with geometric matching quality with the INH in the Tobacco PPO2 crystal and the ACJ in the *M. Xanthus* PPO crystal.^{60,61}

Molecular Dynamics (MD) Simulations. The complexes obtained from docking were used as the starting structures of the wild-type (WT) enzyme, and residue Gly210 was deleted from the WT complexes to produce the mutant-type (MT) structures for the MD simulation. Standard AMBER force field (ff99) was used for amino acid residues,⁶² and the General AMBER force field (gaff) was used for ligands including ligands and cofactor.^{63,64} The partial atomic charges of ligands were determined using the AM1-BCC method,⁶⁵ which was followed by the use of the Antechamber module of the AMBER 8 package. The counterions Na⁺ ions were added to the most

electronegative areas around the protein to neutralize the system, and, then, the whole system was immersed in the truncated octahedron box which extended 8.0 Å away from any solute atoms of TIP3P water molecules.⁶⁶ Energy minimizations and MD simulations were carried out using the Sander module of the AMBER 8 package. In the energy minimization and MD simulations, particle mesh Ewald (PME) was employed to treat the long-range electrostatic interactions.^{67,68} The energy minimization was achieved in two stages. First, movement was allowed only for the water molecules and ions that were minimized with 5000 cycles (2000 cycles of the steepest descent and 3000 cycles of the conjugated gradient minimization). Then, all atoms were energy-minimized with 10 000 cycles (5000 cycles of the steepest descent and 5000 cycles of the conjugated gradient minimization). The SHAKE algorithm was used to constrain all covalent bonds involving hydrogen atoms,⁶⁹ and the time step for MD simulation was set to 2.0 fs. The cutoff distances for the long-range electrostatic and the van der Waals energy terms were set at 10.0 Å. The system was gradually heated from 0 to 300 K over 20 ps, and the equilibrating calculation was executed at a pressure of 1 atm and temperature of 300 K. During the MD simulation process, coordinates were collected every 1 ps for the complexes. A total of 14 model systems, including 6 WT complexes, 6 MT complexes, 1 unbound WT enzyme, and 1 unbound MT enzyme, were simulated in this study (see the Supporting Information for the plots reflecting the MD trajectories). For each MD-simulated complex, 100 snapshots of the structure simulated in the last 200 ps of the stable MD trajectory were used for binding free energy calculations below.

Binding Free Energy Calculation (MM-PBSA). The binding free energies were calculated by using the molecular mechanics–Poisson–Boltzmann surface area (MM-PBSA) binding free energy calculation method. In the MM-PBSA method, the free energy of the receptor/protein–inhibitor binding, ΔG_{bind} , is obtained from the difference between the free energies of the receptor/protein–ligand complex (G_{cpx}) and the unbound receptor/protein (G_{rec}) and ligand (G_{lig}) as follows:

$$\Delta G_{\text{bind}} = G_{\text{cpx}} - (G_{\text{rec}} + G_{\text{lig}}) \quad (1)$$

The binding free energy (ΔG_{bind}) was evaluated as a sum of the changes in the molecular mechanical (MM) gas-phase binding energy (ΔE_{MM}), solvation free energy (ΔG_{sol}), and entropic contribution ($-T\Delta S$):

$$\Delta G_{\text{bind}} = \Delta E_{\text{bind}} - T\Delta S \quad (2)$$

$$\Delta E_{\text{bind}} = \Delta E_{\text{MM}} + \Delta G_{\text{sol}} \quad (3)$$

ΔE_{MM} was calculated by using the following equation

$$\Delta E_{\text{MM}} = \Delta E_{\text{ele}} + \Delta E_{\text{vdw}} \quad (4)$$

where ΔE_{ele} and ΔE_{vdw} are electrostatic and van der Waals interaction energies, respectively, between a ligand and a protein. These energies were computed using the same parameter set as that used in the MD simulation. The solvation free energy ΔG_{sol} consists of two parts:

$$\Delta G_{\text{sol}} = \Delta G_{\text{PB}} + \Delta G_{\text{np}} \quad (5)$$

The electrostatic contribution to the solvation free energy (ΔG_{PB}) was calculated by Poisson–Boltzmann (PB) method using the MM_PBSA module of AMBER 8 program.⁶³ ΔG_{np} is the nonelectrostatic contribution to the solvation free energy determined as a function of the solvent accessible surface area (SASA).^{70,71} Further, the entropic contribution to the binding free energy can be divided into two parts:⁷²

$$\Delta S = \Delta S_{\text{sol}} + \Delta S_{\text{conf}} \quad (6)$$

The ΔS_{sol} is the solvation entropy change, and ΔS_{conf} is the conformational entropy change. Whereas the conformational entropy change is related to the change of the number of rotatable bonds during the binding process, the solvation entropy is gained by the tendency of water molecules to minimize their contacts with hydrophobic groups in protein. We note that the solvation entropy change here could also be considered as an additional term added to the change of the solvation free energy in consideration of the binding process. When an empirical solvation model is developed, one does the parametrization against available experimental data without accounting for the detailed structural information, which means to average the overall solvation contributions from both solvent and solute. Such a mean field approximation is not complete. So, it is appropriate to add the additional term. The computational procedure used to evaluate the entropic contribution ($-T\Delta S$) to the binding free energy was the same as that described in our recent publication.⁷³ As we described previously,⁷³ the solvation entropy change was calculated by using the standard parameters established previously in the literature. The contribution to the binding free energy from the conformational entropy change is proportional to the number (ΔN_{rot}) of the lost rotatable bonds during the binding

$$-T\Delta S_{\text{conf}} = w(\Delta N_{\text{rot}}) \quad (7)$$

in which w is the scaling factor to be calibrated. Hence, our MM-PBSA calculations include an adjustable parameter, i.e., w , in the calculation of the free energy contribution from the conformational entropy

$$\Delta G_{\text{bind}} = \Delta E_{\text{MM}} + \Delta G_{\text{sol}} - T\Delta S_{\text{sol}} + w(\Delta N_{\text{rot}}) \quad (8)$$

although all of the other parameters used in our MM-PBSA calculations are the standard parameters reported in literature or the default parameters of the AMBER 8 program. The specific value of parameter w may be dependent on the type of protein system under consideration. Nevertheless, the adjustment of the w value usually does not change the qualitative order of the calculated binding free energies calculated for a series of compounds binding with a given type of protein.⁷³ This adjustable parameter, w , was calibrated to be 1 kcal/mol here for the PPO proteins. Thus, we used $w = 1$ kcal/mol for our MM-PBSA calculations on all the complexes in the present study. We note that the w value of 1 kcal/mol used in the present study is the same as that used previously by other researchers.⁷⁴

Hydrogen Bond Energy Calculation. The formation of a hydrogen bond is defined in terms of distance and orientation. The combination of donor (D), hydrogen (H), and acceptor (A) with a D–H...A configuration will be considered as a hydrogen bond, when the distance between D and A is shorter than 3.5 Å and the angle D–H...A ranges from 120° to 180°. ⁷⁵ To better

understand the overall strength of hydrogen bonding between the ligand and protein, the hydrogen bond energy (HBE) was calculated using the empirical HBE equation implemented in the AutoDock3.0 program.⁷⁶ The general HBE equation is

$$\text{HBE}(r) \approx 5\epsilon r_{\text{eqm}}^{12}/r^{12} - 6\epsilon r_{\text{eqm}}^{10}/r^{10} \quad (9)$$

in which r is the distance between the donor hydrogen atom (H) and acceptor atom (A), r_{eqm} is the equilibrium internuclear separation between H and A, and ϵ is the energy well depth at r_{eqm} . This HBE equation has recently been used to study the hydrogen-bonding interactions of two cholinesterases with various substrates in the prereactive enzyme–substrate complexes, transition states, and intermediates during the enzymatic reactions,^{77,78} leading to exciting the discovery of high-activity mutants of human butyrylcholinesterase for novel anticocaine therapeutics.^{73,79,80} The parameters r_{eqm} and ϵ for H...O and H...F were determined by performing first-principles electronic structure calculations on the potential energy surfaces (PES in terms of the calculated total energy versus the H...O or H...F distance) of two model systems at the B3LYP/6-31+G**/HF/6-31+G* level (see the Supporting Information for the details). It has been demonstrated^{81–83} that the geometry optimizations at the HF/6-31+G* level followed by the single-point energy calculations at a higher level accounting for electron correlation effects are adequate for various molecules that do not include metal ions. By fitting to the computationally determined PES, we obtained $r_{\text{eqm}} = 1.90$ Å and $\epsilon = 8.4$ kcal/mol for the N–H...O hydrogen bond and $r_{\text{eqm}} = 2.40$ Å and $\epsilon = 2.1$ kcal/mol for the N–H...F hydrogen bond. The HBE values were evaluated for each snapshot taken from the last 200 ps, and the final HBE value was taken as the average for each hydrogen bond.

Results and Discussion

Structures of Proteins and Complexes. We first obtained a three-dimensional (3D) structural model of *A. tuberculosis* PPO2. The 3D model was validated using various protocols. PROCHECK was used to check the stereochemical quality of the homology model. In general, more residues in the most favored regions and less residues in disallowed regions imply a good stereochemical quality of the model. In the homology model, 87.1% of the residues were presented in the most favored regions, 10.7% in the additional regions, 1.9% in the generously allowed regions, and only 0.2% in the disallowed regions (see the Supporting Information). The structural similarity between the homology model and the X-ray crystal structure of tobacco PPO2 was assessed by the values of the root-mean-square deviation (rmsd) of the atomic positions in the modeled structure from those in the X-ray crystal structure of the template. The rmsd of modeled structure from the template structure was found to be 0.94 Å for the α -C (C α) atoms (see the Supporting Information). Further, the inhibitor-binding site of the modeled PPO2 was very close to that of the template. Arg128, the equivalent of Arg98 in the PPO2 of tobacco and Arg95 in the PPO of *M. Xanthus*, is well conserved over all eukaryotic PPOs across species (shown in Figure 1). Leu384 is equivalent to Leu356 in the PPO2 of tobacco and Leu332 in the PPO of *M. Xanthus*. Leu400 is conserved in the PPO2 of tobacco but is replaced by an equally bulky residue, Ile345, in the PPO of *M. Xanthus*. Gly207 and Phe420 are also conserved in many species such as tobacco, but Phe420 is replaced by an equally bulky residue, Met365, in the PPO of *M. Xanthus* (Figure 1).^{60,61}

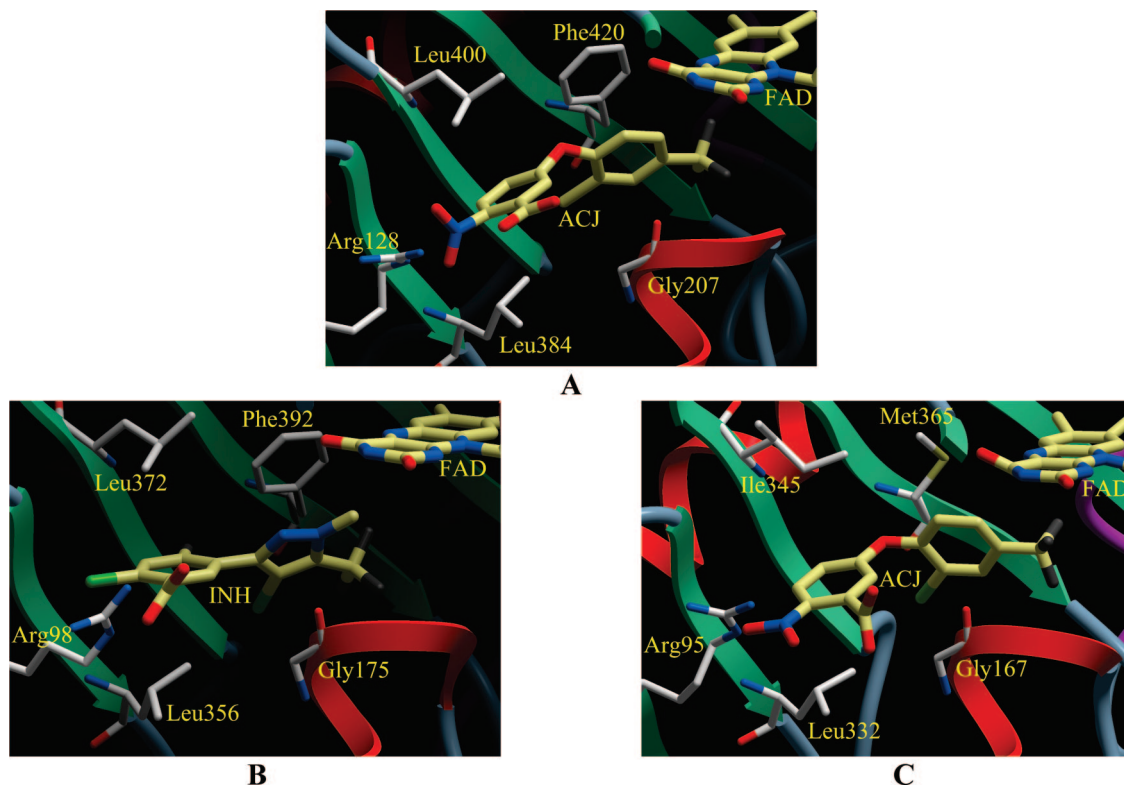


Figure 1. Comparison of the simulated model with the crystal structures: (A) simulated binding mode of *A. tuberculosis* PPO2 in complex with ACJ; (B) crystal structure of tobacco PPO2 in complex with INH; (C) crystal structure of *M. Xanthus* PPO in complex with ACJ.

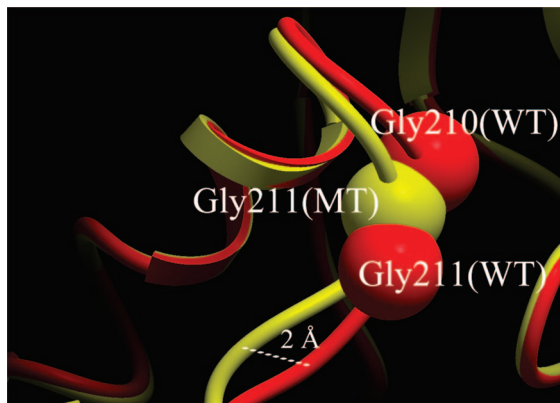


Figure 2. Locations of the residue Gly210 in the WT and MT PPO2. The α carbon atoms of Gly210 and Gly211 are shown as the red CPK model for the WT PPO2 and yellow CPK model for the MT PPO2.

Concerning the 3D structure of the mutant without residue Gly210, as Gly210 lies at the C-terminal end of a helix (Figure 2), its deletion from the wild-type protein structure is not expected to dramatically influence the protein folding. To further test the structure of the mutant, we also constructed a 3D model of the mutant by performing homology modeling with the same template structure as used in the modeling of the wild-type and compared the homology-modeled 3D structure of the mutant to that produced by simply deleting Gly210 in the WT model. The comparison revealed no significant difference between the active site structures of the two types of computational 3D models.

Further, the simulated binding models were also compared with the available X-ray crystal structures of PPO–inhibitor complex. Arg128 in the modeling complex has an ionic interaction with the carboxylate group of the inhibitor ACJ, which is similar to that in the X-ray crystal structure of ACJ in

complex with *M. Xanthus* PPO and the crystal structure of INH in complex with tobacco PPO2 (shown in Figure 1).^{60,61} The carboxylate group substituted phenyl ring of the ACJ is sandwiched between two conserved residues (Leu384 and Leu400), which is also similar to that in the X-ray crystal structures. In addition, there is an aromatic–aromatic interaction between Phe420 and the trifluoromethyl-substituted phenyl ring sandwiched between the conserved residues Phe420 and Gly207, which is also observed in the X-ray crystal structures (Figure 1).^{60,61} Other inhibitors, FOS, LAT, FLC, FLO, and SUT, have very similar binding mode with ACJ and INH because they were thought to inhibit the PPO activity by mimicking the same part of the substrate.³⁰ So, the obtained simulated three-dimensional structure of *A. tuberculosis* PPO2–inhibitor complexes should be reasonable. The detailed results of the sequence alignment and structural evaluations are provided in the Supporting Information.

Effects of Hydrogen Bonding. Hydrogen bonds play an important role in the protein–ligand interactions and make a great contribution to the calculated binding affinity. We have carefully analyzed the effects of hydrogen bond network on the binding affinity. The last 200 snapshots during the equilibrium phase of molecular dynamics simulations were examined, and direct hydrogen-bonding interaction between the inhibitors and protein in each model were identified for both the WT and MT complexes. The numerical results are summarized in Table 1 (see the Supporting Information for more detailed results). In the crystal structure of tobacco PPO2, Arg98, a highly conserved residue in the PPO family, can form a strong hydrogen bond with the ligand.⁶⁰ This hydrogen bond can also be found in the MD-simulated WT complexes of *A. tuberculosis* PPO2.

To better understand the overall strength of hydrogen bonding between the ligand and enzyme, we estimated the hydrogen bond energy (HBE) for each hydrogen bond in each snapshot. In the

TABLE 1: Hydrogen Bond Networks of the WT and MT Complexes

wild-type complexes										mutant complexes						
ligand	acceptor	donor	% ^a	maximum distance ^b	minimum distance	average distance	HBEs ^c	THBE ^d	acceptor	donor	%	maximum distance	minimum distance	average distance	HBEs	THBEs
ACJ	ACJ:O3	Arg128:NE	97.5	3.29	1.70	2.10	-5.4	-9.9	ACJ:O3	Arg128:NE	100.0	2.45	1.66	1.89	-5.1	-8.5
	ACJ:O3	Arg128:NH2	99.5	3.01	1.60	1.91	-4.4		ACJ:O3	Arg128:NH2	81.0	3.49	1.65	2.37	-3.4	
	ACJ:O5	Gly382:N	72.5	4.45	2.55	3.41	-0.2									
FOS	FOS:O3	Arg128:NE	98.5	3.26	1.67	2.09	-5.5	-12.1	FOS:O3	Arg128:NE	78.4	4.53	1.73	2.45	-3.7	-3.7
	FOS:O21	Arg128:NH2	95.8	4.89	1.62	2.04	-5.1									
	Gly207:O	FOS:N2	71.3	4.51	1.77	2.80	-1.5									
LAT	LAT:O1	Arg128:NE	98.3	3.76	1.72	2.07	-6.1	-10.7	LAT:O1	Arg128:NE	86.0	3.83	1.72	2.39	-3.6	-3.6
	LAT:O3	Arg128:NH2	94.0	4.40	1.72	2.24	-4.6									
FLC	FLC:O2	Arg128:NE	100.0	2.48	1.70	1.98	-6.0	-11.5	FLC:F	Thr399:N	96.0	3.88	2.51	3.12	-0.6	-0.6
	FLC:O2	Arg128:NH2	96.0	3.46	1.66	2.12	-5.1									
	FLC:F	Thr399:N	92.0	3.87	2.53	3.22	-0.4									
FLO	FLO:F	Thr399:N	91.5	3.97	2.44	3.24	-0.4	-3.3							0.0	
	FLO:O1	Arg128:NH2	84.5	4.59	1.71	2.82	-1.7									
	FLO:O1	Arg128:NE	78.0	4.52	1.82	2.97	-1.2									
SUT	SUT:O1	Arg128:NE	98.5	2.98	1.68	1.99	-5.6	-10.2	SUT:O1	Arg128:NH1	98.0	2.87	1.69	2.00	-6.2	-6.2
	SUT:O1	Arg128:NH2	95.5	3.37	1.66	2.18	-4.6									

^a Occupancy of hydrogen bonds (the occupancies >70% were listed). ^b Hydrogen bond distance (Å). ^c Hydrogen bond energy (kcal/mol), calculated according to $HBE(r) \approx 5\epsilon r_{eqm}^{12}/r^{12} - 6\epsilon r_{eqm}^{10}/r^{10}$, where the parameters $r_{eqm(H...F)} = 2.40$ Å, $\epsilon(H...F) = 2.1$ kcal/mol, $r_{eqm(H...O)} = 1.90$ Å, and $\epsilon(H...O) = 8.4$ kcal/mol. We calculated the HBE of every snapshot in the last 200 ps and then took the average value. ^d Total hydrogen bond energy (kcal/mol). The total HBE value is the average of the HBE values calculated by using the instantaneous distances in all of the snapshots.

MD-simulated WT complexes, the carbonyl oxygen (see Scheme 1) of each inhibitor formed two N—H...O hydrogen bonds with Arg128 (two side chain nitrogen: NE and NH2). For example, the average HBE value for the hydrogen bond between carbonyl oxygen (O3) of ACJ and the nitrogen (NE) of Arg128 was calculated to be -5.4 kcal/mol, while the corresponding value for the MT complex became -5.1 kcal/mol. The average HBE value between the carbonyl oxygen of ACJ and the nitrogen (NH2) of residue Arg128 was calculated to be -4.4 kcal/mol for the WT complex and -3.4 kcal/mol for the MT complex. As shown in Table 1, compared with that in WT complexes, the calculated hydrogen bond energies between Arg128 and all inhibitors were sharply weakened in the MT complexes. Very interestingly, for the MT complexes of FLC and FLO as shown in Table 1, the hydrogen-bonding interaction with Arg128 disappeared while a new weak hydrogen bond between Thr399 and FLC formed during the MD simulations. As a result, the WT complexes have much stronger overall hydrogen-bonding interactions between inhibitors and the protein than the MT complexes, suggesting that the Gly210 deletion has significant effect on the hydrogen bond interactions between the enzyme and inhibitors.

In order to reveal why a nonactive site residue (Gly210) deletion could result in the remarkable change in hydrogen-bonding interactions between the enzyme and inhibitors, the MD-simulated average structure of the free MT enzyme for the last 200 ps was superimposed with that of the free WT enzyme. Concerning the backbone structure of the enzyme, after Gly210 was deleted, the chain of F204-V205-A206-G207-T208-C209-G211-D212-P213 was contorted and shifted about 2.0 Å as depicted in Figure 2. In addition, in the WT enzyme, Gly210 forms a strong hydrogen bond with the hydroxyl group of the side chain of Ser424 (shown in Figure 3A). The distance between the backbone oxygen atom of Gly210 and the hydroxyl hydrogen atom of Ser424 side chain was about 1.9 Å. The corresponding H-bond distances in the tobacco PPO2 and the *M. Xanthus* PPO crystal structures are 2.3 and 1.8 Å, respectively. According to the results of crystal structures, this interchain hydrogen bond was thought to play an important role in stabilizing the local conformation of the enzyme. Then, as a consequence of the disappearance of this interchain hydrogen bond in the MT enzyme (Figure 3B), the van der Waals interactions between the inhibitors and the binding site residues nearby Gly210 are weakened to different extent (shown in Figure 4).

Binding Free Energies and Drug Resistance. The influence of Gly210 deletion on the binding free energy (ΔG_{bind}) was examined for each modeled complex structure. Summarized in Table 2 are the results obtained from the MM-PBSA calculations for all of the ligands in complex with the WT and MT PPO2. As seen in Table 2, the binding energies (ΔE_{bind} , neglecting the entropic contribution) calculated for all ligands binding with the WT enzyme range from -30.28 to -44.19 kcal/mol. LAT has the lowest ΔE_{bind} value, and SUT has the highest ΔE_{bind} value. However, the ΔE_{bind} values calculated for the MT enzyme binding with the ligands range from -26.05 to -38.40 kcal/mol, with the lowest ΔE_{bind} value corresponding to FLC and the highest ΔE_{bind} value still corresponding to SUT. The calculated entropic contributions ($-T\Delta S$) for each inhibitor binding with both the WT and MT enzymes are close to each other, with values ranging from 9.74 to 11.69 kcal/mol, except for LAT. The calculated $-T\Delta S$ value is 14.37 kcal/mol for the WT LAT-PPO2 complex, but 9.94 kcal/mol for the corresponding MT complex. The Gibbs binding free energies (ΔG_{bind})

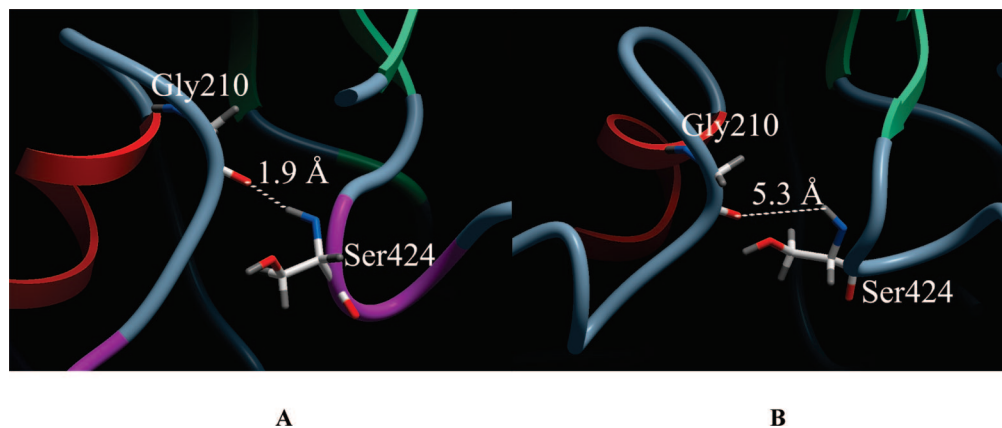


Figure 3. (A) Interchain hydrogen bond between Gly210 and Ser424 in WT PPO. (B) Distance between Gly210 (corresponding to Gly211 in WT enzyme) and Ser424 in MT PPO.

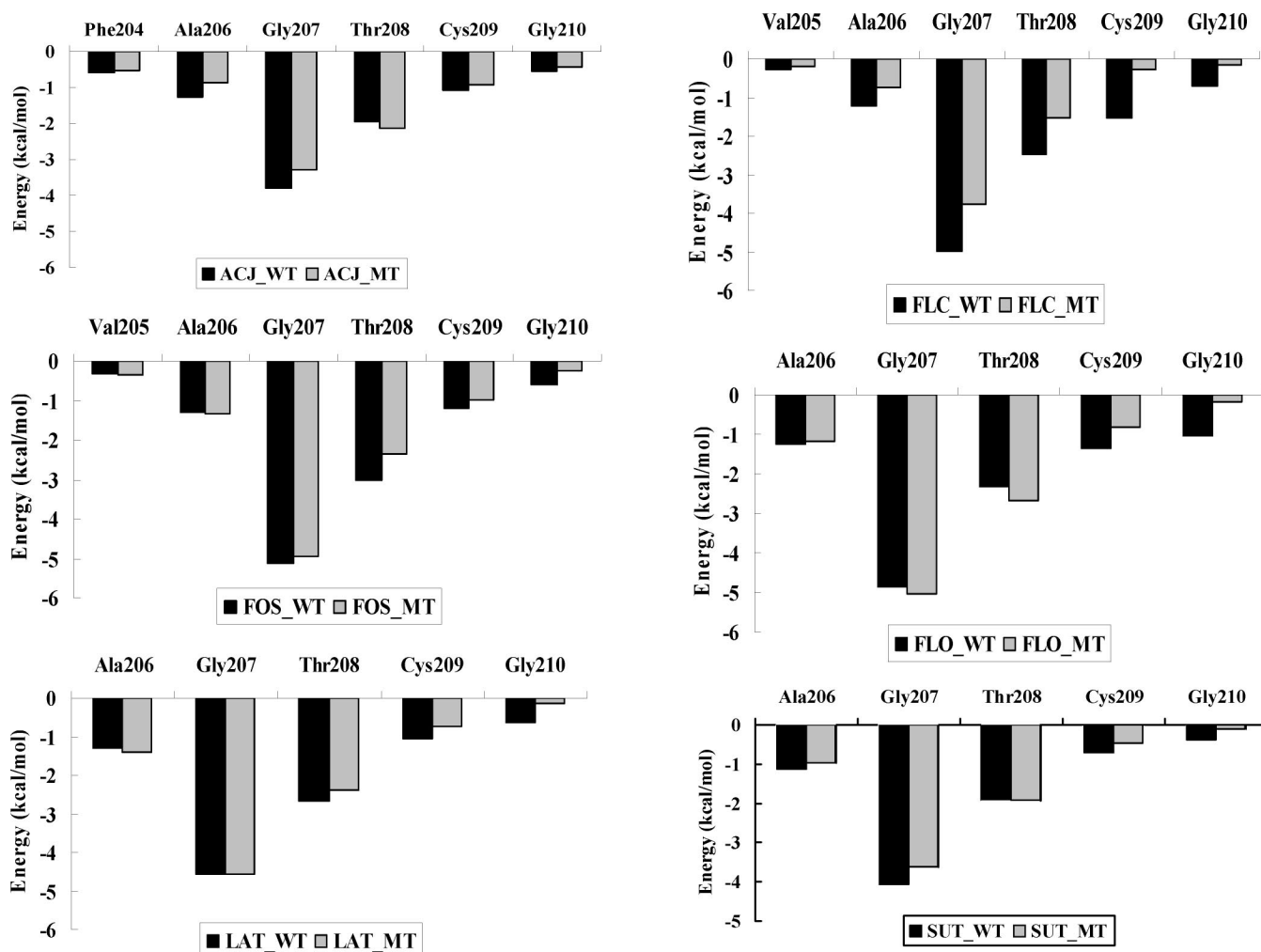


Figure 4. Comparison of the VDW interactions between the inhibitor and local residues around the Gly210 in WT and MT enzyme (in the MT enzyme, the residue Gly211 was renamed Gly210).

calculated for the WT and MT ligand–enzyme complexes range from -16.31 to -29.82 kcal/mol. Obviously, the MM-PBSA calculations most likely overestimated the absolute binding affinities of the ligands with the enzymes. However, the calculated relative ΔG_{bind} values are expected to qualitatively reflect the relative binding affinities of the series of compounds with the enzymes. According to the calculated relative ΔG_{bind} values with the WT enzyme, the highest binding affinity is associated with LAT and the lowest binding affinity is associated with SUT.

The relative binding free energy shifts ($\Delta\Delta G_{\text{bind}}$) from the WT to the MT enzyme calculated for the inhibitors can be used to qualitatively predict the drug resistance levels for the inhibitors due to the mutation (Gly210 deletion). According to the calculated $\Delta\Delta G_{\text{bind}}$ values summarized in Table 2, the Gly210 deletion should cause the drug resistance for all of the inhibitors listed in Table 2 and the highest resistance level should be associated with LAT.

It is important to compare our computational results with available experimental data. Only the in vivo experimental data

TABLE 2: Gibbs Free Energies of Each Model

ligand	$\Delta E_{\text{(binding)}} \text{ (kcal/mol)}$		$\Delta\Delta H \text{ (kcal/mol)}$	$-T\Delta S \text{ (kcal/mol)}$		$-T\Delta\Delta S \text{ (kcal/mol)}$	$\Delta G \text{ (kcal/mol)}$		$\Delta\Delta G \text{ (kcal/mol)}^a$		
	WT	MT		WT	MT		WT	MT	calc.	expt. ^b	RF ^c
ACJ	-39.39	-38.18	1.21	10.50	10.36	-0.14	-28.89	-27.82	1.07	0.47	2.20
FOS	-38.23	-35.70	2.53	10.47	10.76	0.29	-27.76	-24.94	2.82	1.08	6.20
LAT	-44.19	-35.38	8.81	14.37	9.94	-4.43	-29.82	-25.44	4.38	1.85	23.00
FLC	-39.68	-38.40	1.28	10.79	10.63	-0.16	-28.89	-27.77	1.12	0.63	2.90
FLO	-34.30	-30.49	3.81	11.69	10.86	-0.83	-22.61	-19.63	2.98	0.93	4.80
SUT	-30.28	-26.05	4.77	11.03	9.74	-1.29	-19.25	-16.31	2.94	0.97	5.20

^a $\Delta\Delta G = \Delta G_{\text{MT}} - \Delta G_{\text{WT}}$. ^b Obtained according to the equation $\Delta\Delta G_{\text{exp}} = -RT \ln(1/\text{RF})$. ^c Observed resistance factor, defined as $\text{GR}_{50}(\text{MT})/\text{GR}_{50}(\text{WT})$, taken from ref 79.

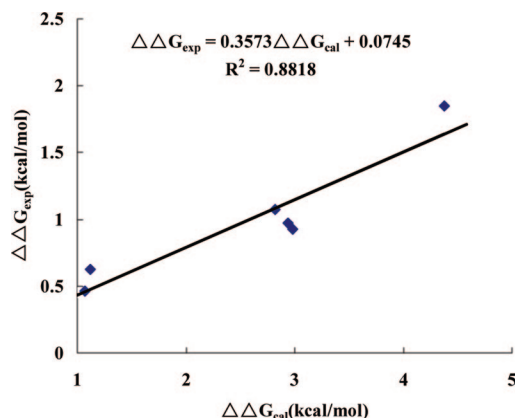


Figure 5. Correlation between the calculated and experimental binding free energy changes.

(GR_{50} , i.e., the growth reduction by 50%) are available for all of the inhibitors listed in Table 2. In principle, the *in vivo* biological activity data are affected not only by the inhibition of the enzyme but also by many other factors that have not been considered in the present study. For this reason, the most potent inhibitor identified *in vitro* is not necessarily the most active compound *in vivo*. Hence, the experimental GR_{50} values should not be compared quantitatively with the calculated enzyme–inhibitor binding affinities. Nevertheless, it is reasonable to assume that the relative *in vivo* activities ($\text{GR}_{50,\text{MT}}/\text{GR}_{50,\text{WT}}$) are usually (although not generally) consistent with the relative binding affinities of the inhibitors with the enzyme. So, in this study, we decided to compare the calculated $\Delta\Delta G_{\text{bind}}$ values with the available experimental data. Qualitatively, all of the calculated $\Delta\Delta G_{\text{bind}}$ values are consistent with the reported experimental results^{84,85} revealing that the Gly210 deletion indeed caused the resistance against all of these inhibitors (listed in Table 2), but it had no impact on the activity of this enzyme. Further, as shown in Figure 5, there is a good linear correlation ($r^2 = 0.88$) between the calculated $\Delta\Delta G_{\text{bind}}$ values and the $\Delta\Delta G_{\text{bind}}$ values derived from the experimental resistance data, $\text{GR}_{50,\text{MT}}/\text{GR}_{50,\text{WT}}$, suggesting that the drug resistance due to the Gly210 deletion is mainly attributed to the decrease in the binding affinity for these inhibitors.

Generally speaking, the binding affinity of an inhibitor to the enzyme is estimated by the Gibbs free energy at the thermodynamic level. According to the equation of $\Delta G = \Delta H - T\Delta S$, three possible mechanisms of drug resistance could be proposed:⁴⁹ Enthalpy and entropy loss (A-type, $\Delta\Delta H$ and $-T\Delta\Delta S$ are both positive, $\Delta\Delta H = \Delta H_{\text{MT}} - \Delta H_{\text{WT}}$, $-T\Delta\Delta S = -T\Delta S_{\text{MT}} - (-T\Delta S_{\text{WT}})$), enthalpy loss with entropy increase (B-type, $\Delta\Delta H$ is positive and $-T\Delta\Delta S$ is negative), or entropy loss with enthalpy increase (C-type, $\Delta\Delta H$ is negative and $-T\Delta\Delta S$ is positive). We note that the calculated ΔE_{bind} in this study can be considered as an approximate result of ΔH . Figure

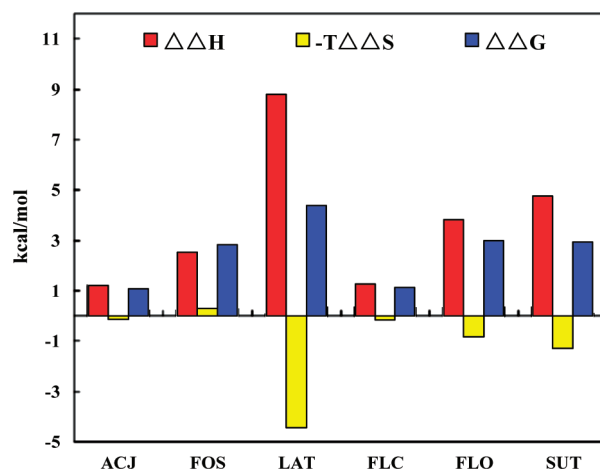


Figure 6. Contribution of entropy and enthalpy changes to the binding free energy changes.

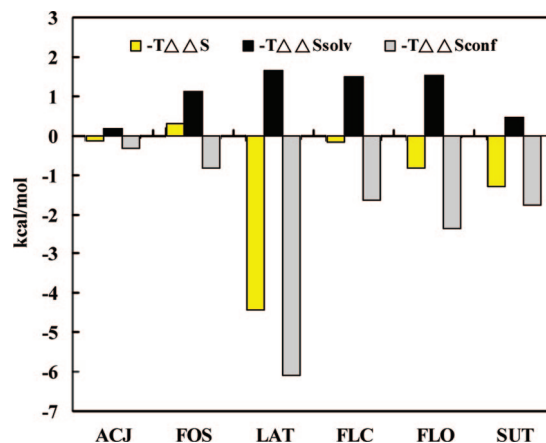


Figure 7. Decomposition of the entropy change.

6 shows the results of the contribution of entropy and enthalpy changes to the binding free energy changes. As shown in Figure 6, compound FOS belongs to A-type mechanism whereas compounds ACJ, LAT, FLC, FLO, and SUT belong to B-type mechanism. Because enthalpy change reflects the strength of hydrogen bonds and van der Waals and other interactions between the inhibitor and enzyme, the positive enthalpy change means that the binding affinity is reduced by the Gly210 deletion whereas the negative enthalpy change means that an improved binding affinity is observed.

It is known that the entropy change consists of two parts: the solvation entropy change ($\Delta\Delta S_{\text{sol}}$) and conformational entropy change ($\Delta\Delta S_{\text{conf}}$).⁷³ Depicted in Figure 7 are the entropy change decomposition of the six inhibitors. As shown in Figure 7, for all of the inhibitors, the fluctuation of the conformational entropy change is always much larger than that of the solvation

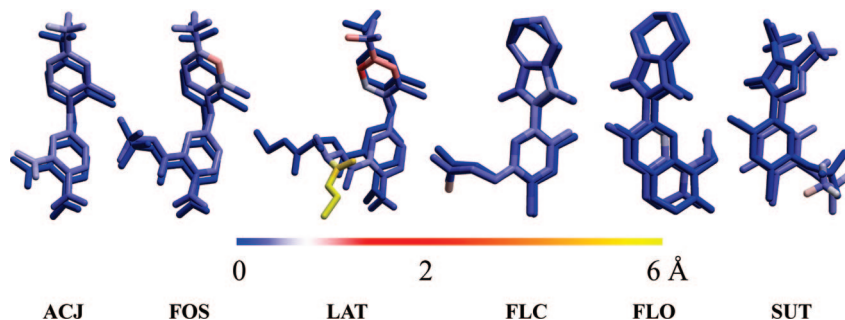


Figure 8. Superimposition of the conformations of inhibitors in the WT and MT models based on a superimposition of the active site. Ligands in the MT model are shown as colored sticks. The ligands in the WT models are shown in deep blue. The color means the magnitude of rmsd shown in the bottom bar.

entropy change. Thus, the conformations of each ligand in the WT and MT enzyme were compared and superimposed as shown in Figure 8. Interestingly, the calculated conformational entropy change seems to relate with the conformational flexibility of inhibitors. The more the conformational flexibility, the more the conformational entropy change. For example, LAT has the largest conformational change of its side chain. Then, the calculated $-T\Delta S$ significantly decreased from 14.37 to 9.94 kcal/mol for LAT. The entropy increase ($-T\Delta\Delta S = -4.43$ kcal/mol) compensated partly the enthalpy loss and reduced the level of drug resistance. It should be noted that LAT experienced the largest enthalpy loss (8.81 kcal/mol) among these six inhibitors; such enthalpy loss should result in over 257-fold resistance if not accounting for the entropy change. That is to say, the entropy increase due to its flexible side chain reduced over 10-fold resistance. Therefore, for the antiresistance inhibitor design, an ideal inhibitor can respond by experiencing very small enthalpy and entropy loss or by compensating enthalpy loss with entropy increase. The more the compensation of the enthalpy loss by the entropy increase, the higher the antiresistant ability an inhibitor should have. In other words, an ideal inhibitor should have enough conformational flexibility to adapt to the conformational change of the binding pocket to form new interactions with the protein.

Conclusion

In *A. tuberculosis* PPO2, as a nonactive site amino acid residue, Gly210 has no direct interaction with the ligands. Thus, it is difficult to speculate the mechanism of drug resistance due to the Gly210 deletion. Our MD simulations and MM-PBSA binding free energy calculations revealed that the MT enzyme (with the Gly210 deletion) has a lower affinity with the ligand than the WT enzyme. The decrease in the binding affinity due to the Gly210 deletion is mainly attributed to the Gly210 deletion-caused disappearance of an important interchain hydrogen bond between Gly210 with Ser424 and the conformational change of the binding pocket. Due to the conformational change of the binding pocket, the favorable hydrogen-bonding interactions between Arg128 of the enzyme and inhibitors are significantly weakened. Our computational results also demonstrate that the calculated conformational entropy change during the binding process seems to relate with the conformational flexibility of the inhibitor in the binding pocket of the MD-simulated enzyme–inhibitor structures. Adaptive inhibitors, which have a flexible molecular structure, are expected to possess the ability needed in the desirable antiresistance inhibitors to inhibit both the wild-type and mutants of a targeted enzyme. The structural and mechanistic insights obtained from

the present study provide a valuable base for rational design of novel inhibitors to overcome the drug resistance associated with the Gly210 deletion.

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Supporting Information Available: Materials about the homology modeling, rmsd graphs during MD simulations of each model, hydrogen bond analysis, and the parameter calculation for the HBE equation. This material is available free of charge via the Internet at <http://pubs.acs.org>.

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